Algebra 1 - Q2.1_2.3 C

Name: ______________________________

2.1 – Expressions and Their Parts

1. I can identify the parts of an algebraic expression

In the following expression, list any **terms**, **coefficients**, and **constants**.

$$-5 + 4a^2 - 3a$$

Terms:

Coefficients:

Constants:

2. I can describe what a variable, constant, coefficient, and term are

Answer each question about the expression $6x - y + 2x^2 + 11$ with **true** or **false**.

a. The expression has 4 terms _______

b. The coefficients are 6, 2, and 11 ______

c. The constant is 11 ______

d. The variables are x, y and z ______

3. I can write a simple expression to model a real-world situation

Your boss says she will pay you $100 to show up to work plus $9 for every hour you work. Write an expression that represents your total pay. Let h be the number of hours you work.

2.2 – Evaluating Expressions

4. I can substitute values into expressions and simplify them

Evaluate the following expressions.

a. $2x - 7$ when $x = -2$

b. $-3a^2 + b - 6ab$ when $a = -4$ and $b = 2$

5. I can use the correct order of operations with variables and parentheses

Evaluate the following expressions.

a. $6(3x - 1)^2 - 10x$ when $x = 2$

b. $-5|y - x| + 3y$ when $x = 8$ and $y = -10$

c. $\frac{1}{x}(x - 10)^2 + x$ when $x = -5$

6. I can use substitution to test whether two expressions are equivalent

Test whether $12x - 1$ is equivalent to $3(4x - 2) + 7$. Try **2 different values** for x.

2.3 – Combining Like Terms

7. I can identify like terms in an expression

For each of the following, indicate whether the terms are like terms (**true** or **false**).

a. $3x$ and $-5y$ _____

b. $-x^2$ and $-12x^2$ _____

c. xy and $-yx$ _____

d. $-2x^2y$ and $9y^2x$ _____

8. I can combine like terms to simplify an expression

Simplify the following expressions by combining like terms.

a. $50 - 2x^2 - 5x + x + 5$

b. $-a + 3b - 5a + 6b^2 - 5 + 2b^2 - 1$

9. I can check my work by substitution or by reasoning about equivalence

Simplify the following expression

$$-2x - 5y^2 + 3x - 15 + 4y^2$$

Test whether your answer is equivalent to the original expression.